Janindu Bandara

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About

- M.Sc. graduate in Electrical Communication Engineering from the **University of Kassel** (defended August 2026, final grade 1.46, degree conferral pending), specializing in *digital signal processing*, *physical-layer waveform design*, and *constrained system optimization*.
- Conducted master's thesis research at **Rohde & Schwarz GmbH & Co. KG**, developing an optimization and validation workflow within an existing **MATLAB** 5G NR link-level simulator to minimize BER for **pre-6G** uplink waveforms across a 630-configuration parameter grid, subject to PAPR, ACLR, and throughput constraints.
- Designed a digital IC in **Verilog HDL** for bisection-based root-finding, implementing **IEEE-754** floating-point arithmetic and a 5th-order Chebyshev polynomial approximation, validated through simulation testbenches; complementary experience programming microcontrollers (ATmega328P, ESP32, NodeMCU ESP8266) and integrating IoT systems.

Experience

Student Assistant (HiWi) — *Kassel, Germany*
University of Kassel — *Sep 2026 – Nov 2026*

- Ported a 630-configuration **MATLAB** simulation campaign to a **Linux SLURM** cluster, parallelizing execution across 24–32 CPU cores and cutting a multi-day sweep to hours.
- Extending the thesis optimization to 3GPP fading channels for a conference publication in preparation.
- **Tech Stack:** *MATLAB (Parallel Computing Toolbox), Linux, SLURM, Bash*

Master's Thesis Student — *München, Germany*
Rohde & Schwarz GmbH & Co. KG — *Oct 2025 – Apr 2026*

- Designed **FDSS** pulse filters with symmetric spectral extension for DFT-s-OFDM waveforms, recovering a ∼2 dB BER penalty caused by per-subcarrier amplitude variation under MMSE equalization.
- Performed constrained waveform optimization over a 630-configuration parameter grid (truncation factor, roll-off, extension ratio), balancing PAPR, ACLR (3GPP Power Class 3), and throughput; identified two operating regimes (energy-preserving at low SNR, PAPR-minimizing at high SNR) across 0–10 dB SNR.
- Benchmarked **FDSS** against CP-OFDM, DFT-s-OFDM, and OTFS across AWGN, TDL-D (3 km/h), and TDL-C (120 km/h) channels; lowest PAPR of the four (∼6 dB below CP-OFDM), ∼2 dB BER penalty vs. DFT-s-OFDM (SE-recoverable).
- **Tech Stack:** *MATLAB (5G NR Toolbox, Communications Toolbox, Signal Processing Toolbox, Parallel Computing Toolbox)*

Temporary Demonstrator — *Colombo, Sri Lanka*
University of Sri Jayewardenepura — *Jan 2022 – Jun 2023*

- Assisted with lab preparation, teaching support, and workshops in the Department of Electrical and Electronic Engineering, covering electronics, communication systems, signal processing, and IoT; ran and graded practical sessions using digital oscilloscopes, function generators, and digital multimeters for circuit and waveform measurement.
- Programmed ATmega328P and ESP32 microcontrollers, simulated logic circuits, and configured routing protocols, VLANs, and DHCP/DNS on Cisco hardware.
- **Tech Stack:** *Arduino UNO, ESP32, Raspberry Pi, Atmel Studio, Proteus, Cisco Packet Tracer, NI Multisim, Wireshark*

Freelance Engineer — *Remote*
Self-Employed (Fiverr) — *Sep 2022 – Oct 2023*

- Developed a web-based IoT visualization interface for real-time sensor data on Google Maps, integrating MQTT and **Node-RED** for live updates.
- Designed a coordinated system integrating a differential-drive mobile robot and a 3-DoF RRR manipulator for autonomous object transfer, implementing GPS-based navigation and inverse kinematics.
- **Tech Stack:** *Node-RED, Google Maps API, JavaScript, MQTT, Docker, Webots, MATLAB (Robotics System Toolbox)*

Telecommunication Engineer Intern — *Colombo, Sri Lanka*
Sri Lanka Telecom PLC — *Feb 2020 – Aug 2020*

- Rotated through Planning, Maintenance, and Network divisions; gained hands-on experience in fiber splicing, OTDR-based verification, and ADSL/FTTH router configuration.
- Supported WDM/MPLS Layer 2/3 VPN provisioning and maintenance of MSAN and eNodeB transmission infrastructure.
- **Tech Stack:** *Fusion Splicer (Sumitomo T-400S), OTDR, MSAN, Clarity OSS, TCP/IP, MPLS, NGN*

Education

University of Kassel — *Kassel, Germany*
Master of Science in Electrical Communication Engineering — *Oct 2023 – Aug 2026*

- **Grade:** 1.46 (degree conferral pending)
- **Thesis:** Pulse Design for Pre-6G Mobile Radio Systems
- **Coursework:** *Mobile Radio, Digital Communications, Signal Detection and Estimation, Information Theory & Coding, Simulation of Digital Communication Systems Using MATLAB*

University of Sri Jayewardenepura — *Colombo, Sri Lanka*
BSc Engineering (Hons) in Electrical and Electronics, minor in Telecommunication Engineering — *Nov 2016 – Jun 2021*

- **Grade:** 3.61/4.0 (Second Class Upper) | equivalent to 1.5 in the German system
- **Thesis:** Versatile Indoor Positioning System using Wi-Fi and Deep Learning
- **Coursework:** *Signal Processing, Digital Communications, Wireless Communication, Embedded Systems, Analog Electronics, Power Electronic Circuit Design, Machine Learning*

Projects

SAE-CNN Based Indoor Positioning System — *Apr 2025 – Aug 2025*

- Reimplemented and evaluated the published CNNLoc framework in **MATLAB** using stacked-autoencoder feature compression and a 1D-CNN; reproduced its building/floor classification performance, measuring 99.73%/95.23% accuracy on UJIIndoorLoc and 92.96% floor accuracy on Tampere.
- **Tech Stack:** *MATLAB (Deep Learning Toolbox), UJIIndoorLoc and Tampere datasets*

Versatile Indoor Positioning System using Wi-Fi and Deep Learning — *Oct 2020 – Jun 2021*

- Fused Wi-Fi RSSI fingerprinting with accelerometer displacement features; KNN ($k = 5$) achieved 72.77 cm mean error (78.7% within 100 cm), up to 35 pp improvement over RSSI-only baseline.
- **Tech Stack:** *Python (scikit-learn, TensorFlow/Keras)*

Root-Finding IC Design — *Nov 2019 – Jan 2020, completed Sep 2026*

- Designed a digital IC for bisection root-finding in **IEEE-754** 16-bit floating-point, featuring a floating-point unit (FPU) and master algorithm unit (MAU) with ln(1+x) approximation via 5th-order Chebyshev polynomial.
- **Tech Stack:** *Verilog HDL, Icarus Verilog*

Technical Skills

Programming Languages: *MATLAB, Python, C/C++, Verilog HDL, Java, JavaScript*

Machine Learning: *scikit-learn, TensorFlow/Keras, MATLAB Deep Learning Toolbox*

Signal Processing: *NumPy, SciPy, Matplotlib; MATLAB 5G NR Toolbox, Communications Toolbox, Signal Processing Toolbox, Parallel Computing Toolbox*

Hardware & EDA: *Icarus Verilog, Vivado, OrCAD, Proteus, Atmel Studio, Arduino IDE*

Databases: *SQLite, MySQL*

Tools: *Git, LaTeX, Docker, Node-RED, Webots, Cisco Packet Tracer, AutoCAD, NI Multisim, Wireshark, Android Studio*

Language Skills

English: Fluent (C1) **German:** B1 (in progress) **Sinhala:** Native